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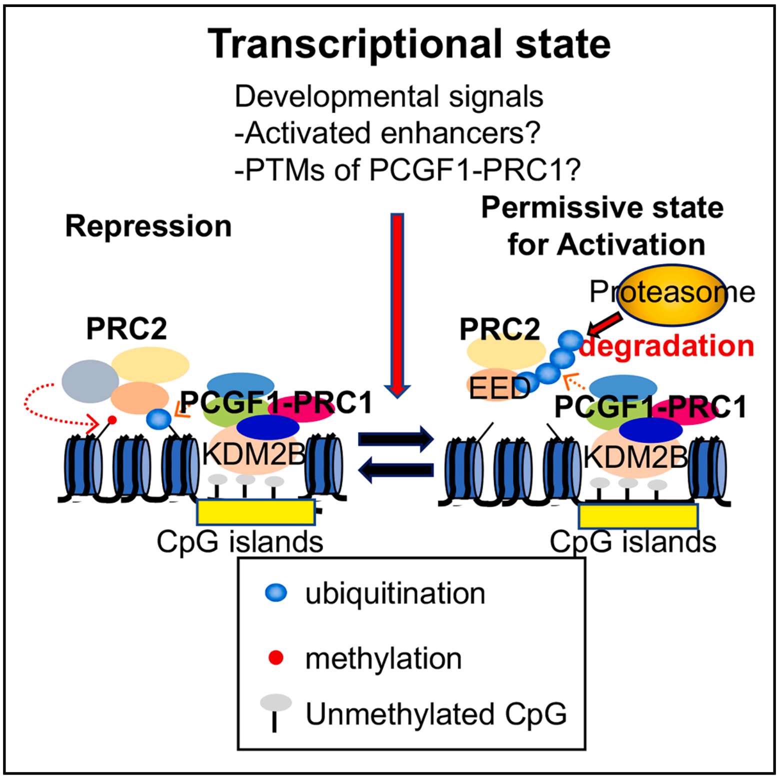
SKP1A bound to Polycomb-silenced genes mediates degradation of PRC2 and preconditions their activation

Graphical abstract



Authors

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In brief

At activation process of silenced developmental genes, Polycomb need to be evicted from the promoter of these genes. Kondo et al. showed that one of Polycomb-repressive complex, PCGF1-PRC1, containing SKP1A-degrade PRC2 component, EED, via ubiquitin-proteasome system for transcriptional activation of developmental genes.

Highlights

- PCGF1-PRC1 associates with SKP1A to regulate the CGI proteome
- SKP1A degrades EED on CGIs to activate repressed genes
- Proteasomal activity is involved in regulation of developmental genes

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SKP1A bound to Polycomb-silenced genes mediates degradation of PRC2 and preconditions their activation

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SUMMARY

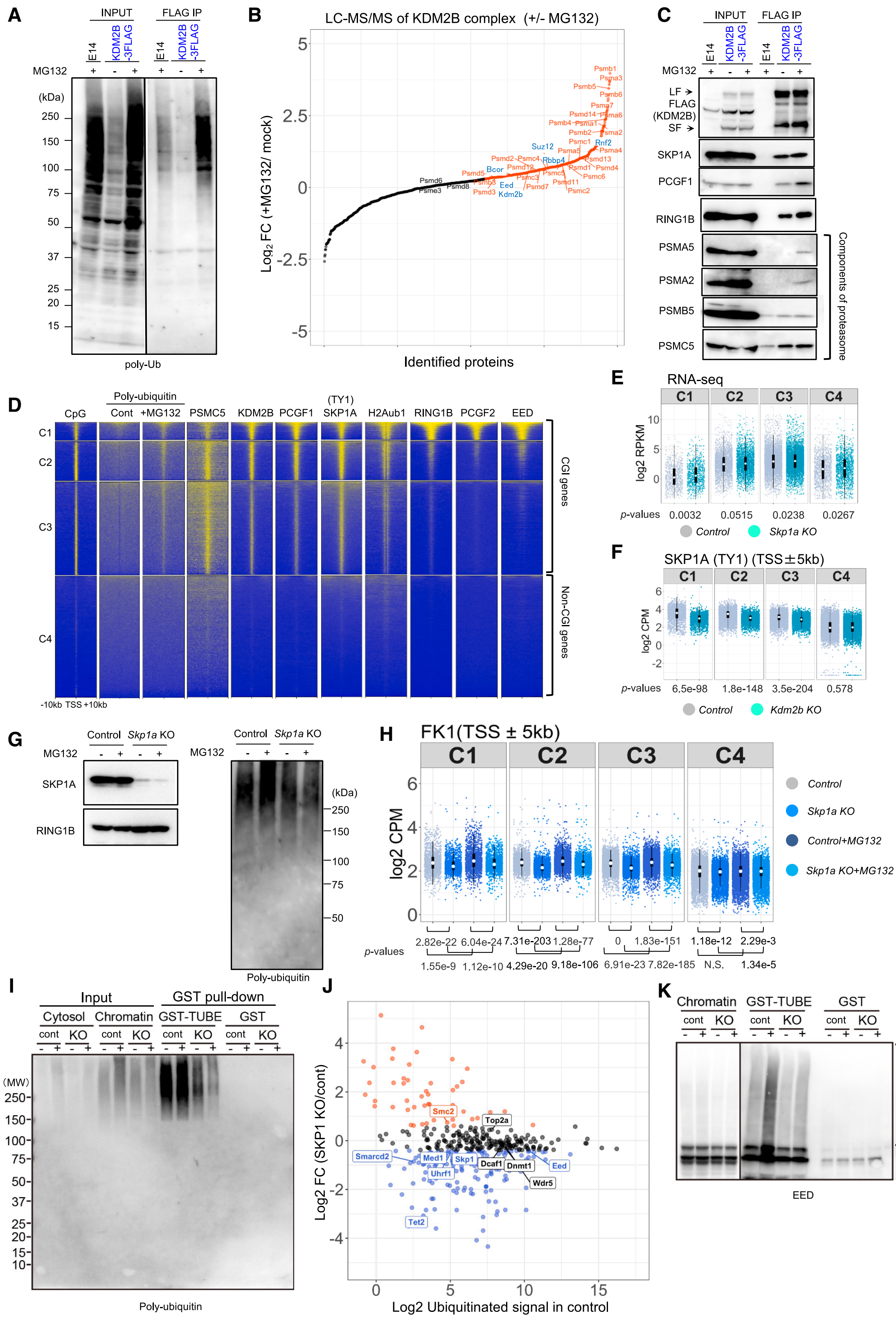
Polycomb group (PcG) proteins are repressors of developmental genes. Paradoxically, the same PcG proteins also function in gene activation via mechanisms that are not yet fully understood. Here, we found that SKP1A, an essential factor of SKP1A/CUL1/F-box (SCF) ubiquitin ligases and Polycomb-repressive complex 1 (PRC1) containing PCGF1 (PCGF1-PRC1), mediates the link between PcG-dependent gene regulation and ubiquitin proteasomal degradation. By using differentiating mouse embryonic stem cells and the midbrain of developing mouse embryos, we found that SKP1A removes EED, a core component of PRC2 that is bound to PcG-target-gene promoters, via proteasomal degradation, and it thereby sensitizes these genes for subsequent activation. In summary, we reveal here a previously unknown role of SKP1A in preconditioning activation of PcG-target genes via the proteasomal degradation of PRC2. This indicates that SKP1A-containing PCGF1-PRC1 may contribute to confer reversibility on PcG-mediated gene silencing for spatiotemporal regulation of target genes.

INTRODUCTION

Precise transcriptional regulation is required for the establishment and maintenance of tissue-specific gene expression during development. Chromatin regulators, such as the Polycomb group (PcG) of proteins, play key roles in cellular development and differentiation, mainly by silencing gene transcription.[1,2] PcG proteins form two distinct complexes: Polycomb-repressive complex 1 (PRC1) and 2 (PRC2). PRC1 mediates mono-ubiquitination of histone H2A at lysine 119 (H2AK119ub1) via RING1A/B[3,4] in concert with PRC2, which mediates H3K27 tri-methylation (H3K27me3).[5,6] Curiously, many developmental genes bound by PcG proteins are reversibly regulated during development.[7] Therefore, PcG proteins, by linking gene promoters with both silencers and enhancers, may reversibly regulate transcription.[8–11]

PcG activity is competitively regulated by trithorax-group proteins that activate transcription.[12,13] Consistent with this notion, PcG proteins promote chromatin compaction, whereas the trithorax-group proteins promote relaxation of the chromatin template *in vitro*.[14,15] It has been hypothesized that the replacement of PcG factors with remodeling factors during gene activation may require the prior removal of PcG proteins from nucleosomes. Indeed, this idea is supported by our previous *in vivo* observation that the activation of the *Meis2* gene during mouse midbrain (MB) development requires the removal of RING1B from the *Meis2* promoter via a mechanism that is paradoxically dependent on PRC1 activity.[8]

PRC1 forms several sub-complexes,[16,17] including variant PCGF1-PRC1 complexes that contain PCGF1, KDM2B, SKP1A, BCOR, PCGF1, RING1A/B, and RYBP. PCGF1-PRC1 recognizes CpG islands (CGIs) via the KDM2B-CXXC domain, and it mediates H2AK119ub1 via the RING1B/PCGF1 dimer.[18–20] H2AK119ub1 facilitates PRC2 recruitment and



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H3K27me3 deposition,[21,22] followed by recruitment of canonical PCGF2/4-PRC1 complexes.[22,23]

In addition to mediating H2AK119ub1, PCGF1-PRC1 is also associated with other ubiquitin-related activities. PCGF1-PRC1 may form SKP1A/CUL1/F-box (SCF) E3 ubiquitin ligase complexes by associating with the F-box of KDM2B.[24–26] PCGF1-PRC1, therefore, may contribute to the ubiquitin-dependent proteasomal degradation system (UPS) during the reversible regulation of target genes.[27–30] Consistent with this notion, chromatin regulation at active genes has been shown to involve the UPS.[29] It is, however, not understood how H2AK119ub1- and UPS-related activities could be differentially regulated by the PCGF1-PRC1 complex. Here, we show that SKP1A, by associating with PCGF1-PRC1, mediates proteasome-dependent eviction of the PRC2 component EED to activate PcG-repressed genes. SKP1A, therefore, connects the H2AK119ub1- and UPS-related activities of the PCGF1-PRC1 complex.

RESULTS

SKP1A regulates polyubiquitination of CGI-bound proteins

To test whether PCGF1-PRC1 is involved in the UPS, we purified KDM2B-associated proteins from mouse embryonic stem cells (ESCs) harboring knockin of a 3×FLAG-tag into the *Kdm2b* locus (E14 + KDM2B-3F ESCs).[19] The ESCs were pre-treated with a proteasome inhibitor, MG132, to capture KDM2B-associated proteins that would otherwise be lost because of rapid proteasomal degradation. Immunoblot analysis with an anti-polyubiquitin antibody (FK1) showed an increase in the ubiquitination level of KDM2B-associated proteins with MG132 treatment, indicating rapid turnover of these proteins via proteasomal degradation (Figure 1A). By mass spectrometric (MS) analysis of proteins co-purified with KDM2B, we identified 674 proteins as KDM2B interactors by excluding non-specific interactors identified by the mock purification. KDM2B showed an increase in its interaction with 26S proteasome components upon MG132 treatment (Figures 1B, S1A, and S1B). This was further confirmed by immunoblotting (Figure 1C). Thus, PCGF1-PRC1 may link polyubiquitinated proteins with the UPS. We also observed that the association of PRC2, PCGF6-PRC1, and chromatin remodelers with KDM2B was also increased by MG132 (Figures 1B, S1A and S1B). In contrast, we did not detect association of KDM2B with CUL1 or RBX1,[26] components of canonical SCF complexes in ESCs, even after MG132 treatment. We therefore further examined this issue by immunoblotting for CUL1 and RBX1 for KDM2B interactors in ESCs and ESCs differentiating into embryoid bodies (EBs) with or without MG132 treatment (Figure S1C). We did not find an association of KDM2B with CUL1 or RBX1, while we did detect its association with RING1B and BCOR. We thus expect that PCGF1-PRC1 may not form canonical SCF complexes.

Because KDM2B directs PCGF1-PRC1 complexes to CGIs via its CXXC domain,[21] we speculated that PCGF1-PRC1 may link CGI-bound proteins with the UPS by polyubiquitinating them. To test this hypothesis, we performed chromatin immunoprecipitation coupled with DNA sequencing (ChIP-seq) and found that polyubiquitinated proteins accumulated at CGIs upon MG132 treatment. In addition, PSMC5, a component of the 26S proteasome, was also constitutively associated with CGIs (Figure 1D).

We next performed a series of ChIP-seq analyses for PcG-related proteins such as PCGF1-PRC1 (KDM2B, PCGF1, and RING1B), PRC2 (EED), and canonical PRC1 (cPRC1) (PCGF2 and RING1B) and H2AK119ub1. PCGF1-PRC1 was associated

Figure 1. SKP1A regulates polyubiquitination of CGI-bound proteins

(A) Polyubiquitination of KDM2B complexes. E14 ESCs expressing 3×FLAG-tagged KDM2B (KDM2B-3FLAG) treated with MG132 were subjected to immunoprecipitation with an anti-FLAG and following immunoblotting with an anti-ubiquitin. Parental E14 ESCs were used for mock purification.

(B) Association of KDM2B with 26S proteasome subunits revealed by immunoprecipitation-mass spectrometry. The label-free quantification intensity of the bait (KDM2B) over the mock control (cont) is plotted with the log ratio between KDM2B purified from MG132-treated versus untreated cells. Peptides that were enriched more than 2-fold are shown in red (proteasome components) and blue (PcG components).

(C) Validation of the immunoprecipitation-mass spectrometry results by immunoblotting for the respective proteins. KDM2B long form (LF) and short form (SF) are indicated.

(D) Accumulation of polyubiquitinated proteins and proteasomal components at CGIs. Heatmaps of CpG density and ChIP-seq enrichment for polyubiquitinated proteins in MG132-untreated (Cont) and -treated (+MG132), PSMC5, PCGF1-PRC1 subunits (KDM2B, SKP1A, and PCGF1), H2Aub1, PRC1 (RING1B and PCGF2), and PRC2 (EED) across TSS regions (±10 kb) in control ESCs. TSSs were clustered into four groups according to binding of KDM2B and RING1B.

(E) Dispensable role of SKP1A for PcG-mediated gene silencing. Dot plot showing the level of gene expression (reads per kilobase per million sequenced reads [RPKM]) in *Skp1a*$^{fl/fl}$ (cont) and *Skp1a*-KO ESCs in each cluster. The Wilcoxon signed-rank test *p* values are shown.

(F) KDM2B-dependent accumulation of SKP1A at CGIs. TY1-tagged SKP1A was expressed in *Kdm2b*$^{fl/fl}$ and *Kdm2b*-KO ESCs. Dot plot shows the SKP1A signals in *Kdm2b*$^{fl/fl}$ (Control) and *Kdm2b*-KO ESCs. CPM, counts per million mapped reads.

(G) Reduction of polyubiquitinated proteins in *Skp1a*-KO ESCs. (Left) SKP1A depletion in *Skp1a*-KO ESCs was confirmed by immunoblotting. (Right) Reduction of polyubiquitinated proteins in *Skp1a*-KO ESCs as revealed by immunoblotting.

(H) SKP1A-dependent and MG132-sensitive accumulation of polyubiquitinated proteins at CGIs. Dot plots show the polyubiquitinated proteins in *Skp1a*$^{fl/fl}$ (Control) and *Skp1a*-KO ESCs untreated or treated with MG132 (+MG132). NS, not significant.

(I) Capture of polyubiquitinated proteins by GST-TUBE. GST pulled down materials from chromatin fractions of *Skp1a*$^{fl/fl}$ (Cont) and *Skp1a*-KO (KO) ESCs by GST-TUBE or GST were immunoblotted with anti-polyubiquitin antibody (FK1). The cells were either treated (+) or not (−) with MG132.

(J) Identification of potential targets of SKP1A-mediated polyubiquitination. MA plot of quantitative LC-MS/MS analysis of polyubiquitinated proteins captured by GST-TUBE revealed a decrease of EED enrichment in *Skp1a*-KO ESCs. The *x* axis shows TUBE-enriched proteins in WT ESCs; the *y* axis shows the signal ratio of TUBE-enriched proteins between *Skp1a*$^{fl/fl}$ (Cont) and *Skp1a*-KO ESCs (n = 2). Red, blue, and black dots indicate proteins with increased, decreased, and unchanged polyubiquitination in *Skp1a*-KO ESCs, respectively.

(K) SKP1A-dependent and MG132-sensitive polyubiquitination of EED. GST pulled down materials from chromatin fractions of *Skp1a*$^{fl/fl}$ (Cont) or *Skp1a*-KO (KO) and untreated (−) or MG132-treated (+) ESCs by GST-TUBE or GST were immunoblotted with anti-EED. The bracket denotes the EED polyubiquitinated signal. See also Figures S1–S3.

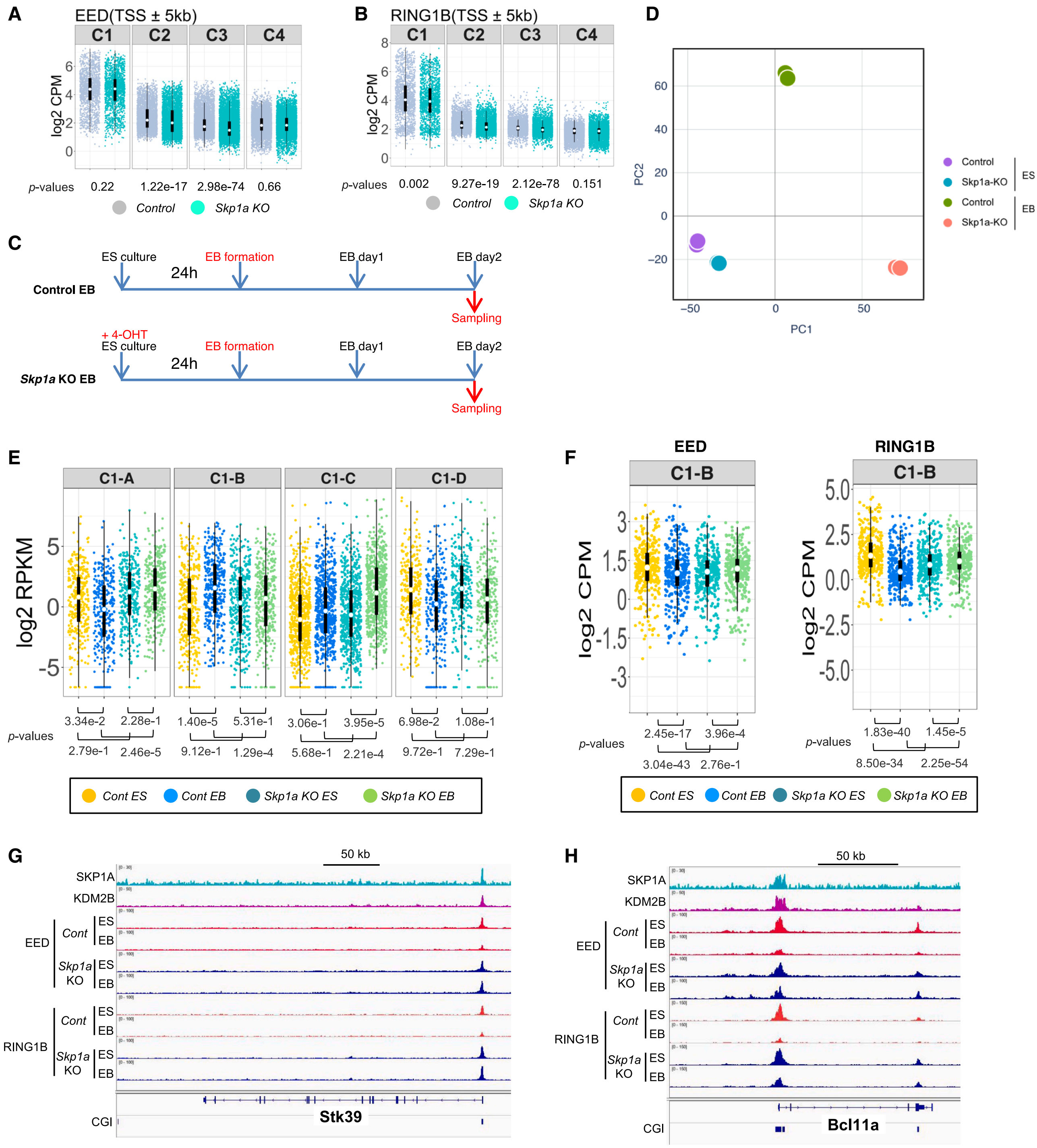

Figure 2. SKP1A triggers the eviction of EED from a fraction of CGIs and potentiates gene activation during ESC differentiation

(A and B) SKP1A is dispensable for target binding of EED and RING1B. Dot plots show the (A) EED and (B) RING1B signals in *Skp1a*$^{fl/fl}$ (control) and *Skp1a*-knockout (KO) ESCs.

(C) Experimental outline of ESC-to-EBs differentiation of *Skp1a*$^{fl/fl}$ (control) and *Skp1a*-KO ESCs.

(D) Involvement of SKP1A in the regulation of gene expression during the ESC-to-EB transition. PCA of RNA-seq data of ESCs and EBs in *Skp1a*$^{fl/fl}$ (control) and *Skp1a*-KO (n = 2).

(E) Role of SKP1A in transcriptional activation of a fraction of PcG-target genes during the ESC-to-EB transition. Dot plots show the level of gene expression of C1 genes. Here, the C1 genes defined in Figure 1D are subdivided into four sub-clusters, namely C1-A–C1-D, according to the gene expression changes during the ESC-to-EB transition. The Wilcoxon signed-rank test p values are shown.

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with most CGI-associated transcription start sites (TSSs), whereas PRC2 and cPRC1 occupied a subgroup of TSSs.

We further classified CGI-associated TSSs into three clusters: C1–C3 (Figure 1D). C1 genes, bound by both PRC2 (EED) and cPRC1, were poorly transcribed, whereas C2 genes exhibited less occupancy by PRC2 or cPRC1 and were more active (Figures 1D and 1E). C3 genes were barely bound by PRC2 or cPRC1, despite the presence of PCGF1-PRC1, but they were the most highly expressed. We defined genes lacking CGIs around their TSS, and without PRC1 or PRC2 binding, as C4 genes. We found that polyubiquitinated proteins exhibited a similar distribution to PCGF1-PRC1 and H2AK119ub1, but not to PRC2 or cPRC1. PSMC5 was enriched around the TSS of C1, C2, and C3 genes.

H2AK119ub1 at PcG-target genes is mediated by RING1A/B, which is associated mainly with PCGF1.[17,31] This indicates that PCGF1 might not play a major role in the polyubiquitination of CGI-bound proteins. Among other components of PCGF1-PRC1, SKP1A, because of its potential association with the UPS, could therefore assume a key role. To test this idea, we exogenously expressed a TY1-tagged SKP1A in an ESC line in which the CXXC domain of KDM2B could be conditionally deleted[21] and found that SKP1A bound to CGIs like KDM2B and PCGF1(Figure 1D). Upon KDM2B disruption, SKP1A binding to CGIs was greatly reduced (Figures 1F and S1D), indicating that the CGI binding of SKP1A requires KDM2B.

We used *Skp1a*-knockout (KO) ESCs to examine whether SKP1A contributes to the polyubiquitination of CGI-associated proteins (Figure S1E). In *Skp1a*-KO ESCs, KDM2B and PCGF1 were modestly reduced, whereas RING1B and H2AK119ub1 levels were unchanged (Figure S1F), indicating that SKP1A does not play a major role in regulating the expression of the canonical functions of PCGF1-PRC1. Nonetheless, the MG132-induced accumulation of polyubiquitinated proteins was reduced in the chromatin fraction in *Skp1a*-KO ESCs (Figure 1G). ChIP-seq analysis with the FK1 antibody showed that MG132-dependent accumulation of polyubiquitinated proteins at CGIs was considerably disrupted in *Skp1a*-KO ESCs (Figures 1H and S1G). We also investigated the level of H2AK119ub1 in the KO ESCs and found that it was modestly downregulated (Figure S1H). Although SKP1A primarily mediates the polyubiquitination of CGI-bound proteins, we speculate that it may play a minor role in the deposition of H2AK119ub1 at CGIs. Together, these results show that SKP1A connects CGI-bound proteins with the UPS.

SKP1A links EED with the UPS

To further elucidate the role of SKP1A in the regulation of CGI-bound proteins, we identified chromatin-associated proteins polyubiquitinated by SKP1A by using the tandem UBA domain of human UBQLN1 (TUBE), which can interact with any type of polyubiquitin chain linkage.[32] We incubated a glutathione S-transferase (GST)TUBE fusion protein with the cross-linked chromatin fraction prepared from wild-type (WT) or *Skp1a*-KO ESCs (Figures S2A and S2B). By immunoblotting with the FK1 antibody, we observed that GST-TUBE pull-down captured polyubiquitinated proteins in control cells, with the level of protein enrichment being enhanced by MG132 treatment, whereas GST alone failed to capture any proteins (Figure 1I). Also, less polyubiquitinated protein was pulled down from *Skp1a*-KO ESCs (Figure 1I). We then used liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis to identify the captured polyubiquitinated proteins by searching for peptides containing the K-e-GG motif (diGly peptides), which indicates the presence of ubiquitination. Out of 177 proteins identified as candidate polyubiquitinated factors, we screened for proteins, such as EED, TET2, SMARCAD2, and MED1, which are known to bind to CGIs or TSSs, or both, and we found EED as a potential candidate that links PcG-mediated gene regulation with the UPS at CGIs (Figure 1J). This was consistent with our earlier finding that the association of PRC2 components (including EED) with KDM2B was increased upon MG132 treatment (Figures 1B, S1A, and S1B). The modest reduction of EED polyubiquitination in *Skp1a*-KO ESCs revealed by the LC-MS/MS analysis was further confirmed by anti-EED immunoblotting (Figure 1K). When we examined the distribution of EED and polyubiquitinated proteins by ChIP-seq, we observed an increase in EED binding and accumulation of polyubiquitinated proteins by MG132 at CGIs of representative PcG-target genes (Figure S2C). Taken together, SKP1A plays a role in mediating polyubiquitination of EED and its subsequent proteasomal degradation.

To examine whether polyubiquitination affects EED stability, we expressed a series of FLAG-tagged mutant EED proteins in which lysine-to-arginine (KR) substitutions were introduced to prevent ubiquitination. EED is reported to be ubiquitinated at multiple sites, in addition to lysine 106 that we identified in this study.[33–36] EED also harbors clustered lysine residues in its N-terminal region. We generated three types of EED mutants harboring KR substitutions in either the N-terminal (N-KR), the C-terminal (C-KR), or the entire (KR) region (Figure S3A). We found that only N-KR could be properly incorporated into PRC2 and restored the H3K27me3 level in *Eed*-KO ESCs (Figures S3B and S3C). We then examined by an *in vivo* ubiquitination assay whether N-KR substitutions dampened EED polyubiquitination. The WT or N-KR version of FLAG-EED was expressed with hemagglutinin (HA)-tagged ubiquitin in 293T cells, with or without MG132 treatment. The FLAG signals of WT EED shifted to a higher-molecular-weight region upon MG132 treatment, demonstrating polyubiquitination. This shift was decreased with the N-KR version (Figure S3D), indicating that EED is polyubiquitinated mainly in the N-terminal region. We compared the binding of WT or N-KR mutant EED at CGIs in ESCs by ChIP-seq. Although the expression level of N-KR EED was lower than that of WT EED, N-KR EED bound to C1 genes more abundantly than did WT (Figures S3B and S3E).

(F) Dot plots of the change in chromatin binding of EED and RING1B in $Skp1a^{fl/fl}$ (Cont) or *Skp1a*-KO ESCs and EBs in cluster C1-B across TSS (±5 kb) regions. The *p* values denote statistical significance, as calculated by the Wilcoxon signed-rank test. NS, not significant.

(G and H) Representative ChIP-seq profiles of EED and RING1B at *Bcl11a* and *Stk39* loci. KDM2B and TY1-tagged SKP1A ChIP-seq tracks are also shown. See also Figure S4.

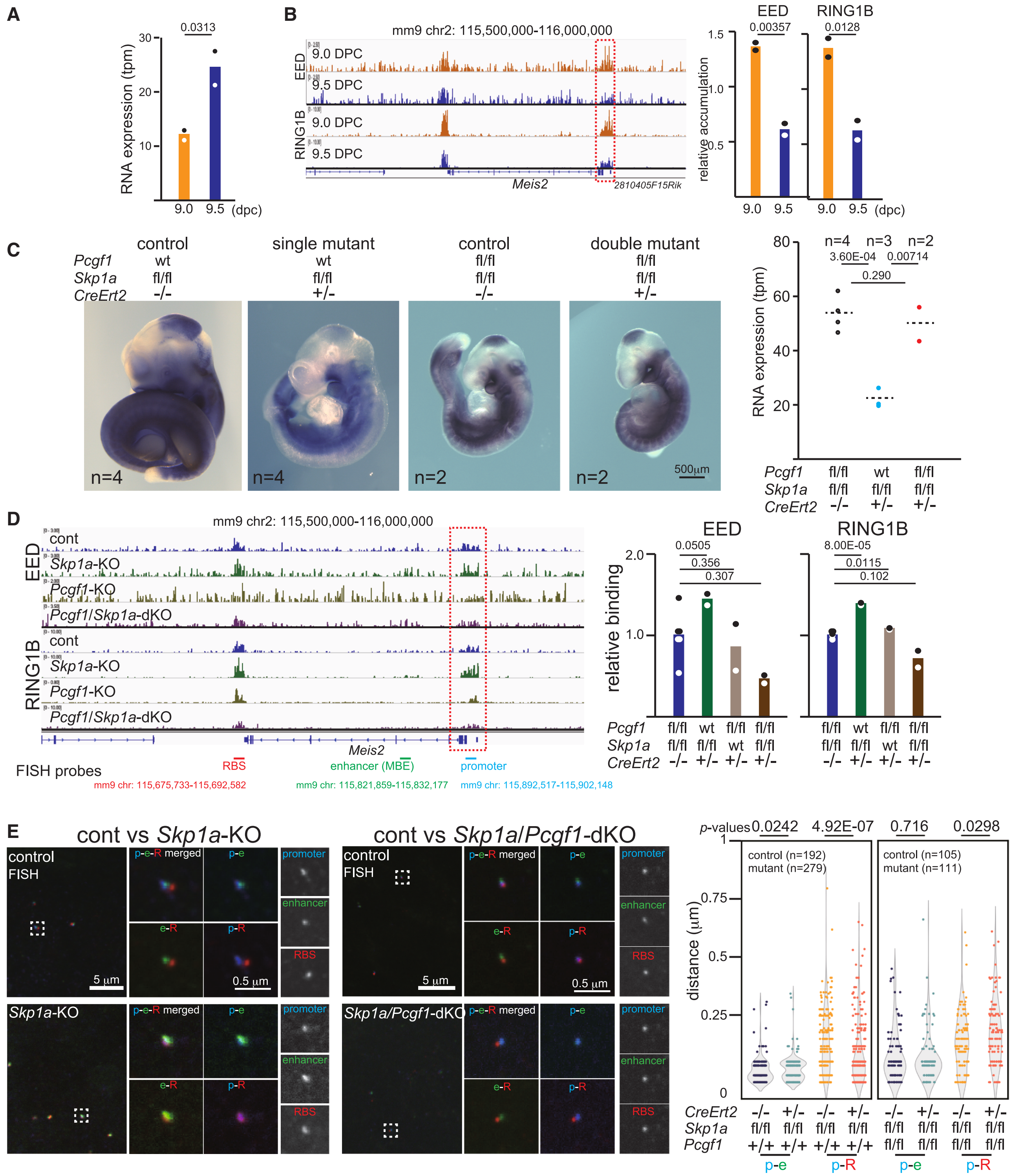

Figure 3. SKP1A is required to evict EED and RING1B from the *Meis2* promoter during murine MB development

(A) *Meis2* upregulation in the midbrain (MB) during the 9.0–9.5 DPC transition. Relative expression levels of *Meis2* in MB of 9.0 and 9.5 DPC embryos revealed by RNA-seq analysis are shown. The *p* value by Student's t test is shown.

(B) Removal of EED and RING1B from the *Meis2* promoter in MB during the 9.0–9.5 DPC transition. (Left) Normalized CUT&Tag signals for EED and RING1B around the *Meis2* locus in MB at 9.0 DPC (orange) and 9.5 DPC (blue). The *Meis2* promoter is indicated by the dotted box. (Right) Normalized cumulative signals for the binding of EED and RING1B at the *Meis2* promoter in MB at 9.0 DPC (orange) and 9.5 DPC (blue). *p* values by Student's t test are shown.

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SKP1A-mediated polyubiquitination may therefore accelerate EED turnover at C1 genes.

SKP1A facilitates EED eviction and transcriptional activation at a group of PcG-target genes during differentiation

Because PCGF1-PRC1 and PRC2 synergistically silence developmental genes in ESCs,[21,37] we speculated that SKP1A-dependent polyubiquitination of EED could be involved in their silencing. In undifferentiated ESCs, *Skp1a* deletion (d2) neither changed the gene expression (Figure 1E) nor perturbed EED or RING1B binding at C1 genes (Figures 2A and 2B). Given that the contribution of SKP1A to EED turnover was limited in undifferentiated ESCs, we wondered whether SKP1A mobilizes EED during ESC differentiation. Supporting this notion, it has been shown that ESC differentiation into EBs induces changes in PcG-target-gene expression.[38,39] SKP1A depletion during the ESC-to-EB transition (Figure 2C) did not cause any obvious morphological changes in either the ESCs or EBs (Figure S4A). Principal-component analysis (PCA) of RNA sequencing (RNA-seq) data showed a similar expression profile in ESCs but a differential expression profile in EBs between WT and *Skp1a*-KO cells (Figure 2D). We found that the C1 genes were dysregulated in *Skp1a*-KO cells during ESC-to-EB transition. We further classified C1 genes into four sub-groups based on variations in transcriptional status alterations during ESC-to-EB transition by *k*-means clustering (Figure 2E). Genes in subgroup C1-B (369 genes) showed transcriptional upregulation together with decreases in EED and RING1B binding (Figures 2E, 2F, S4B, and S4C).

We hypothesized that SKP1A-mediated EED regulation would primarily be active at the genes in the C1-B subgroup. Indeed, transcriptional upregulation and decreased binding of EED and RING1B were dampened in *Skp1a*-KO ESCs (Figures 2E, 2F, S4B, and S4C). Insufficient removal of EED and RING1B in *Skp1a*-KO EBs was confirmed at two representative C1-B genes, namely, *Stk39* and *Bcl11a* (Figures 2G and 2H). SKP1A, therefore, facilitates EED eviction and transcriptional activation at C1-B genes during the ESC-to-EB transition.

We then asked whether polyubiquitination of EED could be involved in facilitating its eviction and transcriptional activation at C1-B genes by using ESCs expressing N-KR EED. Differentiation-associated changes in EED binding and transcription at C1-B genes were comparatively examined between WT and N-KR EED-expressing ESCs. EED eviction at the C1-B genes was more pronounced in WT EED-expressing cells than in N-KR EED (Figure S4D). Accordingly, differentiation-associated upregulation of C1-B genes expression was delayed in N-KR EED-expressing cells compared with WT EED-expressing cells (Figure S4E). We thus suggest that EED polyubiquitination contributes to activation of C1-B genes during the ESC-to-EB transition.

We next wondered why SKP1A-mediated EED eviction occurred specifically at the C1-B genes. Because the C1-B genes were transcriptionally activated during the ESC-to-EB transition, we speculated that SKP1A activity is induced at genes already sensitized for transcriptional upregulation. Consistent with this hypothesis, C1-B genes exhibited the largest increase of the active chromatin mark, H3K27 acetylation (H3K27ac) among the four sub-clusters during the ESC-to-EB transition (Figure S4F). SKP1A-mediated EED eviction is therefore linked with differentiation-associated activation cues.

SKP1A-dependent activation of *Meis2* in the MB coincides with the eviction of EED from the *Meis2* promoter

The ESC-to-EB transition model represents exit from a captured status via the pluripotency network but, because of a lack of defined instructive signals,[38] it does not represent the acquisition of a differentiated phenotype. We therefore investigated the role of SKP1A and PCGF1-PRC1 complexes *in vivo* by focusing on the *Meis2* gene during murine MB development. *Meis2* is bound by RING1B at two separate CGIs around the promoter and the 3′-region (the RING1B-binding site [RBS]) (Figures S5A and S5B).[8] Despite the 200 kb distance between these two CGIs, in the three-dimensional conformation, the RBS is closely associated with the *Meis2* promoter in a RING1B-dependent manner during the transcriptionally inactive state, but it dissociates from the promoter in the active state.[8,40] RBS deletion in ESCs caused the removal of EED and RING1B at the *Meis2* promoter and *Meis2* upregulation (Figure S5B). We have previously shown that priming of *Meis2* expression by the MB enhancer (MBE) accompanies the dissociation of RBS and RING1B from the *Meis2* promoter at 9.0–9.5 DPC (Figure S5A).[8] By performing CUT&Tag,[41] we confirmed that EED and RING1B were removed from the promoter during *Meis2* transcriptional activation (Figures 3A and 3B).

(C) SKP1A-dependent *Meis2* upregulation in MB of 9.5 DPC embryos involves disruption of a PCGF1-mediated pathway. (Left) *Meis2* expression revealed by whole-mount *in situ* hybridization in 9.5 DPC control, *Skp1a*-KO and *Skp1a/Pcgf1*-dKO embryos. Genotypes are shown. (Right) *Meis2* expression revealed by RNA-seq analysis in 9.5 DPC MB of control, *Skp1a*-KO and *Skp1a/Pcgf1*-dKO. TPM values of *Meis2* transcripts revealed by RNA-seq and *p* values by Student's t test are shown.

(D) SKP1A-dependent removal of EED and RING1B from the *Meis2* promoter in MB of 9.5 DPC embryos. (Left) Normalized CUT&Tag signals for EED and RING1B around the *Meis2* gene in MB of 9.5 DPC embryos are shown with their respective genotypes. The *Meis2* promoter region is indicated by the dotted box. The location of the FISH probes for promoter, MB enhancer (MBE), and RBS is indicated. (Right) Average cumulative signals for the binding of EED and RING1B at the *Meis2* promoter in 9.5 DPC MB. Each dot represents a cumulative signal for each experiment. *p* values by Student's t test are shown.

(E) Spatial arrangements of the *Meis2* promoter (p: blue), MBE (e: green), and RBS (R: red) in MB of 9.5-DPC control, *Skp1a*-KO and *Skp1a/Pcgf1*-dKO embryos revealed by FISH analysis. (Left and middle) Low-magnification views of the merged signals, higher magnification views of different combinations of signals and higher magnification views of each signal in *Skp1a*-KO (left) and *Skp1a/Pcgf1*-dKO (middle) with respective controls are shown. (Right) Violin plots for promoter/enhancer (p-e) and promoter/RBS (p-R) distances in MB of 9.5 DPC control, *Skp1a*-KO, and *Skp1a/Pcgf1*-dKO embryos as revealed by FISH analysis. The Wilcoxon test *p* values are shown.

See also Figures S5 and S6.

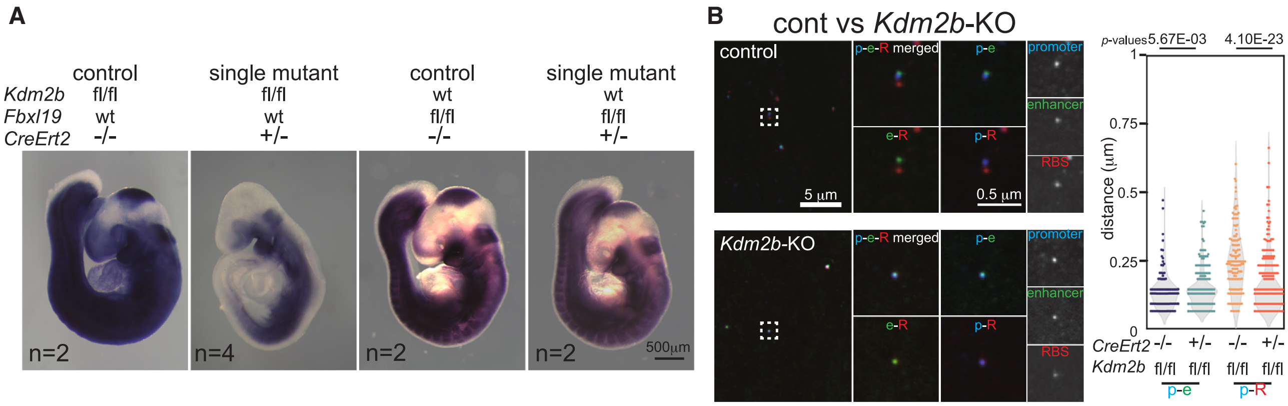

Figure 4. *Kdm2b*-KO mice phenocopy the *Skp1a*-KO

(A) KDM2B-dependent *Meis2* upregulation in 9.5 DPC MB. *Meis2* expression in control, *Kdm2b*-KO and *Fbxl19*-KO at 9.5 DPC is shown.

(B) KDM2B-dependent dissociation of RBS from the *Meis2* promoter. (Left) Spatial arrangement of the *Meis2* promoter (p), MBE (e), and RBS (R) in 9.5 DPC MB of control and *Kdm2b*-KO as revealed by FISH analysis. FISH views in control (top) and *Kdm2b*-KO (bottom) embryos are shown in the same format as those in Figure 3E. (Right) Violin plots of p-e and p-R distances in MB of 9.5 DPC control and *Kdm2b*-KO embryos as revealed by FISH analysis.

See also Figure S6.

To determine the role of SKP1A in the removal of EED and RING1B from the *Meis2* promoter in the MB and subsequent *Meis2* activation, we ablated *Skp1a* (*Skp1a*$^{fl/fl}$) in developing embryos by administering tamoxifen at 8.25 DPC. In *Skp1a*-KO embryos, *Meis2* failed to be activated in MB at 9.5 DPC (Figure 3C). As such a failure of *Meis2* activation in the *Skp1a*-KO could be caused by developmental retardation, we used appropriately developed embryos based on somite number, ranging from 22 to 35, and proper head morphology (Figure S6A). Despite normal development, we observed that *Meis2* activation was disrupted in *Skp1a*-KO embryos. We therefore suggest a role for SKP1A to regulate *Meis2* transcription in developing MB. Furthermore, we found a tendency for EED and RING1B to remain bound to the *Meis2* promoter in the *Skp1a*-KO MB at 9.5 DPC (Figure 3D). In line with this protracted binding of EED and RING1B, RBS dissociation from the *Meis2* promoter was also disrupted in *Skp1a*-KO MB, as observed by FISH (Figure 3E) and 4C-seq (Figure S6B), whereas RBS was dissociated from the promoter in control embryos. Curiously, the interaction between the *Meis2* promoter and MBE was established as normal in the *Skp1a*-KO MB (Figures 3E and S6B). The active chromatin mark H3K27ac was also deposited at the *Meis2* promoter in 9.5 DPC *Skp1a*-KO embryos. This indicated that the *Meis2* promoter was poised in a pre-activation state in the *Skp1a*-KO (Figure S6B, bottom). SKP1A, therefore, is not needed to sensitize the *Meis2* promoter for activation but is required for the removal of EED and RING1B from the *Meis2* promoter and gene activation.

We next investigated whether the protracted binding of EED and RING1B at the *Meis2* promoter disrupts full activation of *Meis2* in MB of *Skp1a*-KO embryos. PCGF1-PRC1 mediates the recruitment of PRC2 and cPRC1[21,38] by depositing H2AK119ub1 and by associating with RING1A/B.[17] We treated *Skp1a*$^{fl/fl}$*Pcgf1*$^{fl/fl}$ double conditional embryos with tamoxifen at 8.25 DPC and found that *Skp1a/Pcgf1* double KO (dKO) led to the restoration of *Meis2* expression in MB (Figure 3C). The protracted association of EED and RING1B at the *Meis2* promoter was diminished, and the failure of RBS dissociation from the *Meis2* promoter was also restored in the dKO (Figures 3D and 3E). Therefore, chromatin defects at the *Meis2* promoter caused by SKP1A depletion were restored by the combined depletion of PCGF1 in MB. Analysis of *Pcgf1*-KO embryos showed that PCGF1 was dispensable for *Meis2* gene activation in MB (Figure S6C). Reduction of EED and RING1B at the *Meis2* promoter, as well as dissociation of RBS from the promoter, also occurred normally in *Pcgf1*-KO embryos (Figures 3D and S6D).

Although KDM2B plays a dominant role in recruiting SKP1A to CGIs (Figures 1D and 1F), other CGI-binding F-box-containing proteins (e.g., KDM2A and FBXL19) may also contribute to this process.[42,43] Consistent with this notion, a role for FBXL19 to link CGIs with poised enhancers via association with the cyclin-dependent kinase module mediator has been reported.[10] In contrast, it has been noted that KDM2A deletion does not overtly perturb PcG-mediated silencing.[44] To determine the role of CGI-binding proteins in *Meis2* transactivation, we conditionally deleted the CXXC domains in KDM2B and FBXL19.[21,42] We found that *Meis2* activation was disrupted in *Kdm2b*-KO embryos but not in *Fbxl19*-KO embryos (Figure 4A). Furthermore, FISH analysis of *Kdm2b*-KO embryos revealed that RBS did not fully dissociate from the *Meis2* promoter, although MBE was normally recruited proximal to the promoter (Figure 4B). The proper recruitment of SKP1A by KDM2B therefore plays a major role in disrupting PcG-mediated *Meis2* silencing in the developing MB.

A molecular link between SKP1A and UPS disrupts *Meis2* silencing

We have shown that SKP1A links EED with UPS-mediated degradation (Figures 2F–2H, S4B, and S4C). We, therefore, examined whether UPS could also be involved in EED eviction from the *Meis2* promoter and gene activation. To this end, we used the proteasome inhibitor bortezomib, as it was less toxic to embryos than MG132. As expected, proteasome inhibition

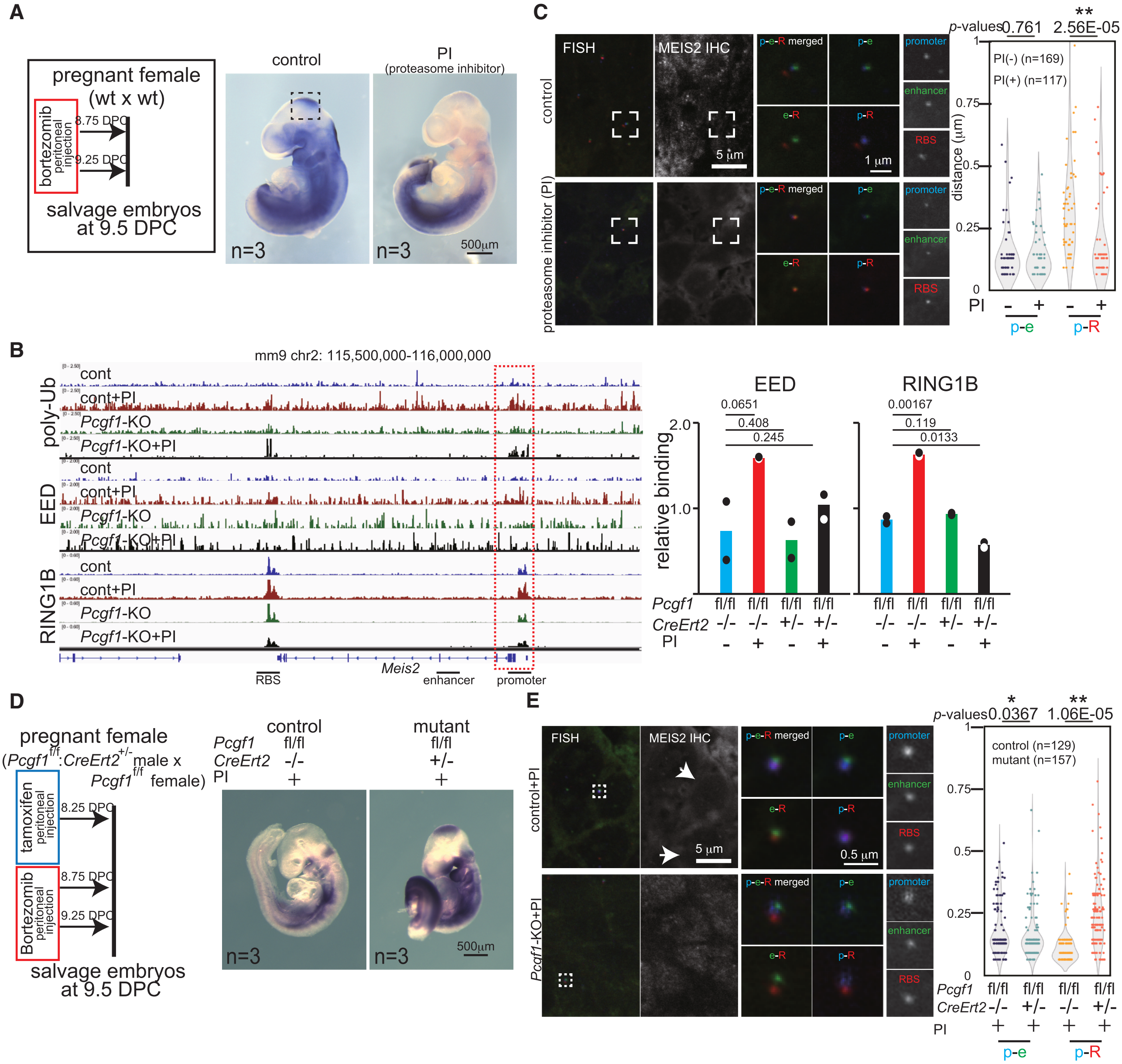

Figure 5. Proteasomal activity is required to evict EED and RING1B from the *Meis2* promoter during murine MB development

(A) Proteasome-dependent expression of *Meis2* in the MB of 9.5 DPC embryos. (Left) Experimental outline of the administration of the proteasome inhibitor (PI), bortezomib. (Right) *Meis2* expression in control and PI-treated embryos at 9.5 DPC.

(B) Proteasome-dependent removal of polyubiquitinated proteins, EED and RING1B, from the *Meis2* promoter in MB of 9.5 DPC embryos. (Left) Normalized CUT&Tag signals for polyubiquitinated proteins, EED and RING1B, around the *Meis2* locus in 9.5 DPC MB of control (cont), PI-treated cont (cont + PI), *Pcgf1*-KO, and PI-treated *Pcgf1*-KO (*Pcgf1*-KO + PI). The *Meis2* promoter region is indicated by the dotted box. (Right) Average cumulative signals for the binding of EED and RING1B at the *Meis2* promoter region in 9.5 DPC MB. Each dot represents a cumulative signal for each experiment. Student's t test *p* values, genotypes and PI treatment are shown.

(C) Proteasome-dependent dissociation of RBS from the *Meis2* promoter in 9.5 DPC MB. (Left) Spatial arrangements of the *Meis2* promoter (p: blue), MBE (e: green), and RBS (R: red) and MEIS2 expression in MB of 9.5 DPC cont and PI-treated embryos as revealed by immunoFISH analysis. In each panel, low-magnification views of the merged signals (FISH) and immunohistochemistry (MEIS2 IHC) images for MEIS2 expression, higher magnification views of boxed regions of different combinations of merged signals, and higher magnification views of each signal are shown. (Right) Violin plots showing promoter/enhancer (p-e) and promoter/RBS (p-R) distances in MB of 9.5 DPC cont (−) and PI-treated (+) embryos. The Wilcoxon test *p* values are shown.

(D) Proteasome-dependent *Meis2* expression in 9.5 DPC MB involves disruption of a PCGF1-mediated pathway. (Left) Experimental outline for PI treatment and *Pcgf1* deletion by tamoxifen administration. (Right) *Meis2* expression in 9.5 DPC PI-treated cont and *Pcgf1*-KO embryos.

(legend continued on next page)

by bortezomib disrupted *Meis2* activation in MB (Figure 5A), accompanied by excess accumulation of polyubiquitinated proteins and protracted binding of EED and RING1B at the *Meis2* promoter (Figure 5B). However, the impact of bortezomib treatment in mid-gestation embryos was variable, and we consequently observed that, depending on the embryo analyzed, 0%–50% of neural cells still expressed MEIS2. We distinguished MEIS2-positive and -negative cells by immunofluorescence analysis and examined the impact of bortezomib treatment in each group. In MEIS2-negative cells, we observed protracted RBS association with the *Meis2* promoter and normal promoter-MBE interactions (Figure 5C). These results indicate a role for SKP1A-dependent UPS in the eviction of EED and RING1B, and dissociation of RBS from the *Meis2* promoter, after MBE recruitment.

To test whether proteasome activity facilitates *Meis2* activation by disrupting PcG-mediated silencing, we deleted *Pcgf1* in bortezomib-treated embryos. Strikingly, the failure of *Meis2* activation induced by bortezomib in the MB was restored in *Pcgf1*-KO (Figure 5D). Bortezomib-induced protracted binding of EED and RING1B at the *Meis2* promoter and failure of RBS dissociation were also rectified in *Pcgf1*-KO embryos (Figures 5B and 5E). UPS, therefore, plays a crucial role in activating *Meis2* by disrupting PcG-mediated silencing in MB. Given the role of SKP1A in EED polyubiquitination, as observed in ESCs, we propose that SKP1A links EED with UPS-mediated degradation to remove PRC2 from the *Meis2* promoter and activate gene expression in the developing MB.

SKP1A-dependent removal of EED is a prerequisite for upregulation of target genes during MB development

We finally asked whether SKP1A-mediated removal of EED could function downstream of activation cues or in parallel with them for upregulation of PcG-repressed genes in the developing MB. We thus asked to what extent changes in EED binding in *Skp1a*-KO MB correlated with transcriptional changes of EED-bound genes. We found 2,344 genes bound by EED by combining CUT&Tag data from 9.0 and 9.5 DPC MB. EED binding at these genes was compared between the control and *Skp1a*-KO MB at 9.5 DPC. We unexpectedly observed a modest but general gain of EED binding at these target genes in the *Skp1a*-KO (Figure 6A). SKP1A therefore contributes to widely removing EED from target genes in 9.5 DPC MB. We next examined whether SKP1A-mediated removal of EED was accompanied by transcriptional upregulation of target genes by integrating changes in transcription levels of EED-bound genes with those in EED binding levels in the *Skp1a*-KO (Figure 6A). Here, EED-bound genes were arbitrarily classified into five clusters (clusters 1–5 in MB: C1M, C2M, C3M, C4M, and C5M) (Figure 6A). With this classification, we found that genes that gained EED binding tended to be downregulated in the *Skp1a*-KO, as represented by 660 genes downregulated (C1M) whereas 340 genes were upregulated (C3M). However, this analysis also revealed that a considerable number of EED-bound genes were still upregulated in the *Skp1a*-KO. Importantly, we found that the C1M and C3M genes were bound by similar amounts of EED and modestly acquired EED binding upon SKP1A depletion in a PCGF1-dependent manner (Figure 6B). Therefore, SKP1A similarly regulates target binding of EED at the C1M and C3M genes, nonetheless, the transcriptional outcomes diverge. We, therefore, expect that SKP1A-mediated regulation of target binding of EED does not solely determine transcriptional outcomes of the target genes. Instead, SKP1A-dependent EED eviction may contribute to the preconditioning of EED-bound genes for forthcoming activation in the developing MB. We thus speculate that SKP1A-mediated removal of EED is a prerequisite for the subsequent activation of EED-bound genes and may cooperate with other determinants for their transcriptional upregulation.

DISCUSSION

In this study, we showed that SKP1A binds to CGI-containing TSSs via interaction with KDM2B, which in turn promotes EED degradation via the UPS. We demonstrated this role of SKP1A in EED degradation in two distinct developmental contexts, namely, the ESC-to-EB transition and MB development. In the ESC-to-EB transition, SKP1A activity removes EED at a subgroup of PcG-bound genes that are sensitized for transcriptional activation. In the developing MB, SKP1A removes EED from the *Meis2* promoter in a UPS-dependent manner. This removal is essential for *Meis2* activation after its priming by interaction with MBE.

We expect that PCGF1-PRC1 includes two paradoxical functions for the regulation of PcG-mediated gene silencing via the differential usage of either RING1A/B or SKP1A. RING1A/B mediates stable gene silencing via H2AK119ub1 deposition and subsequent PRC2 recruitment, whereas SKP1A contributes to the disruption of PcG-mediated gene silencing by degrading EED (Figure 6C). Our study indicates that SKP1A activity could be primed by receipt of development- or differentiation-associated activation cues. SKP1A removes EED from genes sensitized for activation, as illustrated by the C1B genes in the ESC-to-EB transition (Figure 2F), and from *Meis2* following MBE recruitment in developing MB (Figure 3D). In support of this model, previous reports have shown that post-translational modification, such as phosphorylation of SCF E3 ligases or substrate proteins, or both, could alter the affinity between E3 and substrates and thus ubiquitinate the substrates.[26,45] Such differentiation-associated activation cues could conditionally alter the interaction between KDM2B/SKP1A and EED to trigger polyubiquitination of EED. Importantly, our results also indicate that SKP1A-mediated removal of EED is a prerequisite, but not sufficient, to disrupt PcG-mediated gene silencing, as revealed in the MB (Figures 6A and 6B). We thus expect a role for SKP1A to

(E) Proteasome-dependent dissociation of RBS from the *Meis2* promoter in 9.5 DPC MB involves disruption of the PCGF1-mediated pathway. (Left) ImmunoFISH images for PI-treated cont and *Pcgf1*-KO embryos are shown in the same format as those presented in (C). MEIS2-negative cells are indicated by arrows. (Right) Violin plots showing promoter/enhancer (p-e) and promoter/RBS (p-R) distances in MB of 9.5 DPC PI-treated cont and *Pcgf1*-KO embryos. The Wilcoxon test *p* values are shown.
See also Figure S6.

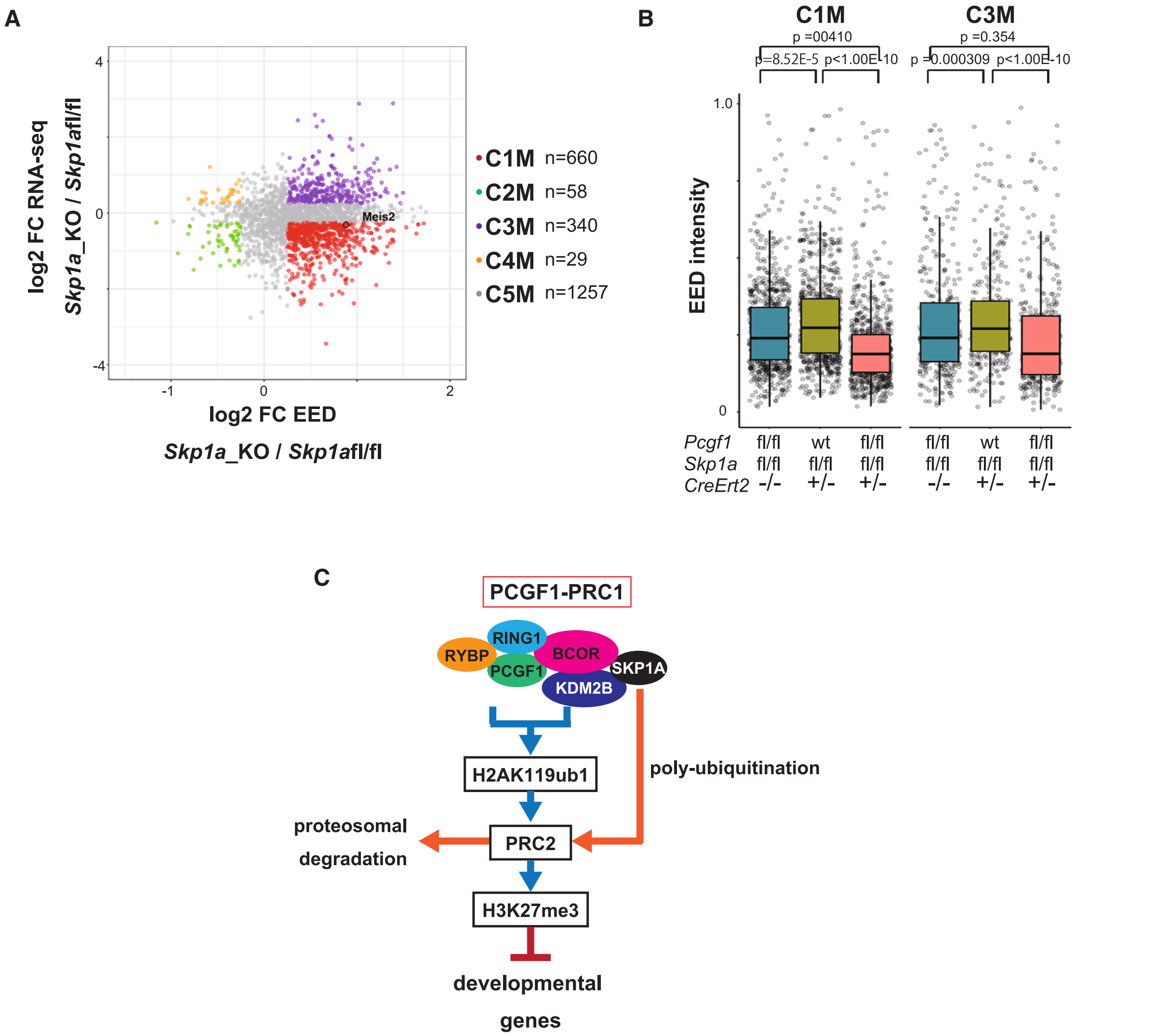

Figure 6. Genome-wide analyses of SKP1A-dependent transcriptional activation

(A) Classification of PcG-target genes into five clusters by combining RNA-seq and CUT&Tag data from cont and *Skp1a*-KO MB. *Meis2* is included in cluster 1 in MB (C1M).

(B) Scatterplots showing EED binding at each gene in the C1M and C3M cluster found in cont, *Skp1a*-KO, and *Skp1a*/*Pcgf1*-dKO MB. The *p* values by the HSD of ANOVA-Tukey are shown.

(C) Schematic summary of the role for SKP1A in disrupting PCGF1-PRC1-initiated gene silencing.

See also Figure S6.

precondition EED-bound genes for forthcoming activation by removing EED upon receipt of differentiation-associated activation cues.

Our results also indicate that KDM2B is involved in the activation of *Meis2* in the MB, suggesting its role in SKP1A-mediated EED eviction (Figures 4A and 4B). KDM2B is expected to link SKP1A with ubiquitination substrates via its interaction with leucine-rich repeats (LRRs), which often contribute to the capture of ubiquitination substrates.[46–48] Intriguingly, although several studies have reported an association of KDM2B with CUL1/RBX1, which are core components of the canonical SCF ligase, to degrade substrate proteins that associate with the LRRs of KDM2B,[26,49] we and others have failed to detect CUL1/RBX1 in KDM2B complexes (Figure S1C).[18,24,43,46] These paradoxical observations may imply that KDM2B/SKP1A forms a bona fide SCF ligase complex only under certain conditions and that it degrades target proteins, as was shown in the removal of EED from target genes after the receipt of activation cues (Figures 3D and 3E). This model, however, needs expanding to accommodate the previously accepted role of LRRs in associating with BCOR or its paralog BCORL1 in the PCGF1-PRC1 complex.[50,51] *Bcor* mutation has been shown to lead to a spectrum of hematopoietic abnormalities reminiscent of those induced by *Pcgf1*-KO, and PUFD domain of BCOR were shown interacting with PCGF1 through RAWUL domain[50,52–56] to stabilize ternary interactions involving KDM2B.[50] We therefore assume that the LRRs of KDM2B could be pre-occupied by BCOR or BCORL1 incorporated into the PCGF1-PRC1 at PcG-silenced genes to avoid the premature association of SKP1A with its substrates. This scenario also implies that dissociation of BCOR or BCORL1 from the LRRs could be a prerequisite to bringing substrates such as EED within the proximity of SKP1A for their future eviction. Therefore, the LRRs of KDM2B could be an alternative platform for facilitating either PRC2 recruitment by bringing RING1A/B via PCGF1/BCOR or by PRC2 removal by SKP1A-mediated polyubiquitination at CGIs. However, how could this binary decision be made by the LRRs of KDM2B? BCOR is reported to associate with not only PCGF1-PRC1 but also other chromatin regulators such as BCL6 and MLL3T.[57,58] Therefore, BCOR/KDM2B association could be regulated by such chromatin regulators that can interact with BCOR. Whatever the underlying mechanisms, these apparently paradoxical roles of the PCGF1-PRC1 complex seem to comprise a mechanism for maintaining robust, yet reversible, gene repression.

The role of the UPS in the regulation of gene transcription is well established in eukaryotes.[29,30] Subunits of the 26S proteasome are recruited to the GAL10 gene locus in an activation-dependent manner in yeast.[59] Accumulating evidence, including in mammalian cells, has also established the critical role of proteasome-mediated degradation in optimizing the amount of transcription factors bound to chromatin.[30,60–63] However, how proteasomes associate with the chromatin during gene activation remains poorly understood. Our results indicate that CGIs can serve as a platform for UPS-mediated chromatin-bound protein regulation via SKP1A-dependent ubiquitination followed by degradation via the 26S proteasome complex. Intriguingly, we show that PSMC5, a component of the 26S proteasome complex, constitutively associates with CGIs regardless of their transcription status, indicating that part of the 26S proteasome is preloaded at CGIs. Importantly, KDM2B, KDM2A, and FBXL19 contain the CXXC domain and an F-box. Although those proteins associate with CGIs,[18–20,42,64] the changes caused by their respective depletion in ESCs are not necessarily the same. Differential usage of F-box proteins could be a part of the mechanism underlying the substrate specificity of SKP1A at CGIs. It is noteworthy that about 70 F-box proteins that may accommodate SKP1A have been identified in mammals.[65] Although, unlike KDM2A/B and FBXL19, many of them lack a CXXC, the participation of other F-box proteins cannot be formally excluded.[42,43,64] Future studies are needed to examine this possibility.

In summary, this study provides the first evidence of a SKP1A-mediated link between PcG-mediated gene silencing and UPS-mediated protein degradation. Moving forward, it will be important to elucidate the molecular mechanism that reinforces SKP1A recruitment by PCGF1-PRC1 and triggers the polyubiquitination of EED for the subsequent transcriptional activation step.

Limitations of the study

In this study, we proposed the role for SKP1A to link UPS with PcG-target CGIs via its association with KDM2B and facilitate EED eviction to mediate differentiation-associated activation of PcG-silenced genes. This conclusion, however, could be dampened by a lack of substantial evidence showing the contribution of KDM2B-bound SKP1A to mediate EED eviction. In theory, this issue could be addressed by using F-box mutant of KDM2B, which can form complexes with PCGF1, RING1B, and BCOR but not SKP1A. Nonetheless, amino acid residues that contribute to link the F-box of KDM2B with SKP1A turned out to also play a role in the formation of PCGF1-PRC1. This point needs further clarification to consolidate the model. Another issue that requires further investigation is to identify E3 ligases that are active in polyubiquitination of EED. In ESCs or EBs, KDM2B was revealed to not form complexes with either CUL1 or RBX1. We thus expect that KDM2B-associated SKP1A may contribute to form atypical SCF complexes. This hypothesis needs further experimental clarification.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to the lead contact Haruhiko Koseki (haruhiko.koseki@riken.jp).

Materials availability

Reagents generated in this study will be shared by the lead contact upon request.

Data and code availability

All proteomics data can be accessed from the ProteomeXchange Consortium via jPOST[66]: PRIDE: PXD047552 (KDM2B IP-MS) and PXD047556 (GST-TUBE pull-down). GEO accession numbers of the NGS data including RNA-seq, ChIP-seq, and CUT&Tag are GEO: GSE263260, GSE263261, GSE263263, GSE263265, GSE293161, GSE293160, GSE297093, GSE297094, and GSE297309.

This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

H.K., T.K., and S.I. conceived the project and designed the experiments. T.K., S.I., Y.-W.H., K.K., M.S., Y.K., and M.N. performed the experiments. J.T., T.E., M.H., H.S., and O.O. analyzed the data. All of the authors interpreted the data. H.K., T.K., S.I., and J.S. wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

- ○ Cell culture
- ○ Mice
- ● METHOD DETAILS
  - ○ Generation of the conditional mutant allele of *Skp1a*
  - ○ FLAG immunoprecipitation for mass spectrometric analysis
  - ○ Sample preparation of proteins co-purified with KDM2B for LC-MS/MS measurement
  - ○ Sample preparation of GST-TUBE pull-down captured proteins for LC-MS/MS measurement
  - ○ RNA-seq
  - ○ ChIP-seq and CUT&Tag
  - ○ ChIP-seq and CUT&Tag data processes
  - ○ FISH and immunoFISH
  - ○ 4C-seq
- ● QUANTIFICATION AND STATISTICAL ANALYSIS
  - ○ Analyses for LC-MS/MS measurement
  - ○ The analyses for ChIP-seq and CUT&Tag
  - ○ The analysis of FISH

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti-RING1B	Kondo et al.[8]	N/A
Rabbit polyclonal anti-PCGF1	Blackledge et al.[21]	N/A
Mouse monoclonal anti-PCGF1	Santa Cruz	sc-515371; RRID: AB_2721914
Rabbit monoclonal anti-SUZ12	CST	3737; RRID:AB_2190850
Rabbit polyclonal anti-PCGF2	Sugishita et al.[38]	N/A
Rabbit monoclonal anti-H3K27me3	CST	9733; RRID: AB_2616029
Rabbit monoclonal anti-H2AK119ub1	CST	8240
Mouse monoclonal anti-H3K27ac	Millipore	8173
Mouse monoclonal anti-BCOR	Santa Cruz	sc-514576; RRID: AB_2721913
Rabbit polyclonal anti-BCOR	Proteintech	12107-1-AP
Rabbit polyclonal anti-KDM2B	Blackledge et al.[21]	N/A
Rabbit polyclonal anti-RYBP	Millipore	AB3637; RRID: AB_11213333
Rabbit monoclonal anti-SKP1A	CST	2156
Rabbit monoclonal anti-HA	CST	3724; RRID: AB_1549585
Rabbit polyclonal anti-PSMC5	Enzo	BML-PW8895; RRID: AB_10540901
Rabbit polyclonal anti-PSMA5	CST	2457
Rabbit polyclonal anti-PSMA2	CST	2455; RRID: AB_2171400
Rabbit monoclonal anti-PSMB5	CST	12919; RRID: AB_2798061
Mouse monoclonal anti-polyUb (FK1)	MBL	D071-3; RRID: AB_592938
Rabbit polyclonal anti-FLAG	Sigma	F7425; RRID: AB_439687
Mouse monoclonal anti-TY1	Sigma	SAB4800032
M2-anti-FLAG resin	Sigma	A2220; RRID: AB_10063035
Goat polyclonal anti-Mouse IgM	Millipore	AP128P; RRID: AB_92481
Mouse monoclonal anti-MEIS2	Abcam	ab55647
Rabbit monoclonal anti-CUL1	Abcam	Ab75817
Rabbit monoclonal anti-RBX1	CST	11922
Rabbit monoclonal anti-RING1B	CST	5694
Rabbit monoclonal anti-EED	CST	85322
Rabbit monoclonal anti-KDM2B	CST	44570
Chemicals, peptides, and recombinant proteins		
3×Flag peptide	Sigma	F4799
anti-FLAG-M2 affinity gel agarose	Sigma	A2220
Protein A/G Magnetic Beads	Pierce	88803
4-hydroxytamoxifen	Sigma	H7904
Alexa Fluor 488-5-dUTP	Thermo Fisher	C11397
Cy5-dCTP	GE Healthcare	PA55021
Cy3-dCTP	GE Healthcare	PA53021
Critical commercial assays		
RNeasy Mini Kit	Qiagen	74104
NEBNext poly(A) mRNA magnetic isolation module	NEB	E3370
NEBNext Ultra II RNA library preparation kit for Illumina	NEB	E7775
Library quantification kit	Takara	638324
NEBNext Ultra II FS DNA prep kit	NEB	E7805

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Raw and analyzed data	This paper	GEO: GSE263260
Raw and analyzed data	This paper	GEO: GSE263261
Raw and analyzed data	This paper	GEO: GSE263263
Raw and analyzed data	This paper	GEO: GSE263265
Raw and analyzed data	This paper	GEO: GSE293160
Raw and analyzed data	This paper	GEO: GSE293161
Raw and analyzed data	This paper	GEO: GSE263260
Immunoblot and CBB data	This paper; Mendeley Data	https://data.mendeley.com/preview/hd89kyjb5n?a=abfe4337-841d-4719-9dfe-ab24d30e0218
Figure 3C_Skp1a_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/dsw2f7r32y?a=6dd9c6bf-7be2-4404-aa86-13b5efa596d1
Figure 3C_Skp1aPcgf1_dKO_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/9vrphrc5x4?a=2e6e8a26-a792-4645-8a18-0848b4d7c65e
Figure 3E_Skp1a_FISH	This paper; Mendeley Data	https://data.mendeley.com/preview/pd78d3nwjv?a=62b7a02a-f634-402f-a375-501913c9dfa1
Figure 3E_Skp1a/Pcgf1_dKO_FISH	This paper; Mendeley Data	https://data.mendeley.com/preview/nnw3yg55pn?a=b75cd729-0db0-44ad-968b-fcb50c500836
Figure 4A_Fbxl19_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/k2sxm9bck5?a=f6c0ba38-8409-42bb-88e0-cf5a15ee07f1
Figure 4A_Kdm2b_FISH	This paper; Mendeley Data	https://data.mendeley.com/preview/wgsz8st48c?a=01dde760-817e-4545-bba3-bd179179ec90
Figure 4B_Kdm2b_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/p8842jxx8n?a=882f51eb-57c7-45bd-9341-d46fb71ce080
Figure 5A_PI_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/fvgzb27krc?a=9095e3a4-9531-49a2-9bd1-84ff09d051f3
Figure 5C_PI_ImFISH	This paper; Mendeley Data	https://data.mendeley.com/preview/m5vjmrddgf?a=d5859abf-16cc-4d2f-abea-16167f43a50e
Figure 5D_PIPcgf1_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/5n365grymv?a=4f51f4e2-0c28-408d-9488-7d88054923c7
Figure 5E_PIPcgf1_ImFISH	This paper; Mendeley Data	https://data.mendeley.com/preview/jyfvj8mntn?a=e786e7fd-bd58-479a-986b-032466f4083e
Figure S6C_Pcgf1_wish	This paper; Mendeley Data	https://data.mendeley.com/preview/wtw6xr7dj3?a=142e0353-381a-4877-abcc-a7ee1319a39f
Figure S6D_Pcgf1_FISH	This paper; Mendeley Data	https://data.mendeley.com/preview/h4prb634v2?a=6ff2d0ce-948a-40f2-8eed-328dab20d4de
Experimental models: Cell lines		
Mouse: Skp1a$^{fl/fl}$ ESCs	This paper	N/A
Mouse: Kdm2b$^{fl/fl}$ ESCs	Blackledge et al.[21]	N/A

(*Continued on next page*)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mouse: Pcgf1$^{fl/fl}$ ESCs	Sugishita et al.[38]	N/A
Mouse: Eed$^{fl/fl}$ ESCs	Sugishita et al.[38]	N/A
Mouse: E14+KDM2B-3F ESCs	He et al.[19]	N/A
Experimental models: Organisms/strains		
Mouse: Skp1afl/fl: B6-Skp1a$^{tm1.1HK}$/J	This paper	N/A
Mouse: Pcf1fl/fl: B6-Pcgf1$^{tm1.1HK}$/J	Sugishita et al.[38]	N/A
Mouse: Kdm2bfl/fl: B6-Kdm2b$^{tm1.1HKTK}$/J	Blackledge et al.[21]	N/A
Mouse: Fbxl19fl/fl: B6-Fbxl19$^{tm1.1HK}$/J	Dimitrova et al.[42]	N/A
Oligonucleotides		
See Table S1 for oligonucleotides	This paper	N/A
Software and algorithms		
bowtie2	Langmead and Salzberg[67]	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Samtools	Li et al.[68]	N/A
Picardtools		http://broadinstitute.Github.io/picard
deepTools2	Ramírez et al.[69]	https://deeptools.readthedocs.io/en/latest/index.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture

Mouse embryonic stem cells (ESCs) were cultured in D-MEM with 20% fetal bovine serum, MEM nonessential amino acids (Gibco), L-glutamine (Gibco), 2-mercaptoethanol (Gibco), and Leukemia Inhibitory Factor (LIF) on irradiated primary mouse embryonic fibroblast feeder layers at 37°C in a 5% CO_2 incubator. For induction of conditional knockout, 800 nM (final concentration) of 4-hydroxytamoxifen (Sigma) dissolved in ethanol was added to the culture medium. To inhibit proteasome activity, ESCs were treated with the reversible proteasome inhibitor MG132 (Sigma) (10 μM) for 3 h before analysis. For embryoid body (EB) formation, ESCs were cultured without LIF for 2 days on low attachment culture dishes.

Mice

Mice were housed under specific-pathogen-free conditions in accordance with the regulations of the RIKEN animal facility. Conditional knockout mice were backcrossed with C57BL6N background mice several times and then crossed with mice harboring the *ROSA26^{CreERT2}* locus.[70] For knockout, 0.1 mL of 15 mg/mL tamoxifen (T5648; Sigma-Aldrich) dissolved in corn oil (C8267; Sigma-Aldrich) was injected intraperitoneally into pregnant females at 8.5 DPC. Then, 0.17 mL of 0.1 mg/mL bortezomib (504314; Sigma-Aldrich) dissolved in DMSO (D2650; Sigma-Aldrich) was intraperitoneally injected into pregnant females at 8.75 DPC, and 0.1 mL of 0.1 mg/mL bortezomib was injected intraperitoneally into pregnant females at 9.25 DPC. Mouse embryos from mutants used in this study were stage with 22 to 35 somites considered as over 9.5 DPC.

METHOD DETAILS

Generation of the conditional mutant allele of *Skp1a*

Generation of *Pcgf1* flox (*Pcgf1^{fl}*), *Kdm2b* flox (*Kdm2b^{fl}*), and *Eed* flox (*Eed^{fl}*) alleles has been described previously.[21,71] A targeting vector to generate the *Skp1a* flox allele (*Skp1a^{fl}*) was generated by modifying genomic regions derived from bacterial artificial chromosome clone (RP23-440M17) using the double Red recombination method as described previously.[72] In this targeting vector, parallel *loxP* sites were inserted such that they flanked exon 2 of *Skp1a*, which encodes an ATG initiation codon and part of the POZ domain (Figure S1E). The linearized targeting vector was introduced into mouse ESCs (laboratory-made ESCs originated from hybrid embryos derived from C57BL/6J and 129Sv mice) by electroporation (GenePulser; Bio-Rad). The ESCs were grown on feeder cells in an appropriate medium. Each colony was collected and expanded, and the genomic DNA of each clone was purified. To identify true homologous recombinant colonies, direct sequencing was conducted by polymerase chain reaction (PCR) amplification using the outward primers for the homologous region and the inward primers for the neomycin resistance gene. Targeted ESCs were used to generate chimeric mice. The resulting chimeric mice were backcrossed with 57BL/6J mice for more than five generations.

FLAG immunoprecipitation for mass spectrometric analysis

KDM2B complexes were purified from E14 mouse ESCs in which a sequence for 3×Flag-tag was knocked into the carboxy terminus of the *Kdm2b* locus (E14 +KDM2B-3F cells). Cells were harvested by scraping, washed with 1× phosphate-buffered saline (PBS), and equilibrated in four pellet volumes of Nonidet P-40 (NP-40) lysis buffer (NETN) (50 mM Tris-HCl [pH 7.9], 150 mM NaCl, 1% NP-40, 1 mM EDTA [pH 8.0]). The cells were incubated on ice for 10 min and then sonicated briefly, and whole-cell lysates were recovered by centrifugation at 15,000 rpm for 15 min. FLAG-tagged proteins were immunoprecipitated from the lysates by using anti-FLAG-M2 affinity gel agarose for 4h at 4°C, with rotation. The agarose beads were washed five times with NETN buffer, then the bound materials were eluted twice with 60 µL of NETN buffer containing 0.2 mg/mL of 3×Flag peptide (Sigma).

Sample preparation of proteins co-purified with KDM2B for LC-MS/MS measurement

Sample preparation of proteins co-purified with KDM2B as shown in Figures 1B and S1A for liquid chromatography-tandem mass spectrometry (LC-MS/MS) measurement was performed as reported previously.[73] Samples were treated with 10 mM dithiothreitol at 50° C for 30 min and then subjected to alkylation with 30 mM iodoacetamide in the dark at room temperature. The mixture was diluted four-fold with 50 mM ammonium bicarbonate and digested by using 800 ng Lys-C and 400 ng trypsin overnight at 37°C. An equal volume of ethyl acetate was added to the digested sample, and the mixture was acidified with 0.5% trifluoroacetic acid (final concentration). The mixture was shaken for 5 min and centrifuged at 22,000 rpm for 5 min for phase separation; then, the aqueous phase was retrieved. The volume of the recovered digested sample was reduced to at least half of the original volume by a centrifugal evaporator for the complete removal of ethyl acetate and then the mixture was desalted by using C18-Stage Tips. The peptides trapped in the C18-Stage Tips were eluted with 40 µL of 50% acetonitrile and 0.1 % trifluoroacetic acid, and this was followed by drying using a centrifugal evaporator. The dried peptides were dissolved in 20 µL of 3% acetonitrile and 0.1% formic acid, then 2 µL of the sample was analyzed by LC-MS/MS.

A preliminary data-dependent acquisition was performed for SWATH protein quantification, as described below.[73] Peptides (approx. 100 ng) were directly injected onto a 100 µm × 15 cm PicoFrit emitter (New Objective) packed in-house with 120-Å porous C18 particles (ReproSil-Pur C18-AQ 1.9 µm; Dr. Maisch GmbH; pC18_B) and then separated by using a 240-min acetonitrile gradient (3%–40%; flow rate, 300 nL/min) and an Eksigent Ekspert nanoLC 400 HPLC system (Sciex). Peptides eluting from the column were analyzed on a TripleTOF 5600+ mass spectrometer. MS1 spectra were collected in the range of 400–1200 *m/z* for 250 ms. The top 25 precursor ions with charge states of 2+ to 5+ that exceeded 150 counts per second were selected for fragmentation with rolling collision energy, and MS2 spectra were collected for 100 ms. A spray voltage of 2100 V was applied.

All MS/MS files were searched against the UniProtKB/Swiss-Prot mouse database (Proteome ID: UP000000589, downloaded 11 September 2019, 17,019 protein entries), combined with the standard MaxQuant contaminants database, by using the ProteinPilot software v. 4.5 and Paragon algorithm for protein identification. The search parameters were as follows: cysteine alkylation of iodoacetamide, trypsin digestion and TripleTOF 5600. For the protein confidence threshold, we used the ProteinPilot unused score of 1.3 with at least one peptide with 95% confidence. Moreover, proteins and peptides identified as contaminating proteins were excluded. The global false discovery rate for both peptides and proteins was less than 1%.

SWATH data-independent acquisition was performed by using the same gradient profile as used for the data-dependent acquisition experiments described above. Precursor ion selection was done in the 400–1200 *m/z* range, with a variable window width strategy (7–75 Da). The collision energy for each individual SWATH experiment was set at 45 eV, and 80 consecutive SWATH experiments (100–1800 *m/z*) were performed, each lasting 36 ms.

Sample preparation of GST-TUBE pull-down captured proteins for LC-MS/MS measurement

The samples of GST-TUBE pull-down captured proteins as shown in Figure 1J were prepared as described in a previous study for DIA-MS analysis.[74] In brief, the samples were digested by trypsin and Lys-C mix (Promega) after reduction and alkylation of cysteine residues using tris (2-carboxyethyl)phosphine and iodoacetamide.

The digested peptides were directly injected onto a 75 µm × 35 cm, PicoFrit emitter packed in-house with 2.7 µm core shell C18 particles at 50°C and then separated with a 120-min gradient (A = 0.1% formic acid in water, B = 0.1% formic acid in 80% acetonitrile) consisting of 0 min 10% B, 90 min 36% B, 110 min 60% B, 120 min 60% B at a flow rate of 100 nL/min using an UltiMate 3000 RSLCnano LC system (Thermo Fisher Scientific, Waltham, MA, USA). Peptides eluting from the column were analyzed on an Orbitrap Exploris 480 MS (Thermo Fisher Scientific) with an InSpIon system[75] for overlapping window DIA-MS.[76,77] MS1 spectra were collected in the range of *m/z* 495-925 at 15,000 resolution to set an automatic gain control target of 2e6 and a maximum injection time of "auto". MS2 spectra were collected in the range of *m/z* 200-1800 at 30,000 resolution to set an automatic gain control target of 3e6, maximum injection time of "auto", and stepped normalized collision energy of 22, 26, and 30%. An isolation width for MS2 was set to 6 m/z and overlapping window patterns in 500-920 m/z were used as window placements optimized by Skyline software.[78]

MS files were searched against a Mouse UniProtKB/Swiss-Prot database using Scaffold DIA (Proteome Software). The Scaffold DIA search parameters were as follows: experimental Data Search ()enzyme, trypsin; maximum missed cleavage sites, 3; precursor mass tolerance, 9 ppm; fragment mass tolerance, 9 ppm; fixed modification, carbamidomethylation [C]; variable modification, ubiquitination [K]. The peptide identification threshold was set at peptide false discovery rates of less than 1%.

RNA-seq

Total RNA was extracted by using an RNeasy Mini Kit (Qiagen). Poly(A) mRNA was isolated from total RNA by using an NEBNext poly (A) mRNA magnetic isolation module (New England Biolabs), and then a library was prepared by using an NEBNext Ultra II RNA library preparation kit for Illumina (New England Biolabs). The average size of each library was measured with an Agilent 4200 TapeStation (Agilent Technologies). The size of each library was measured with a Library quantification kit (Takara). Sequences were read by a NextSeq500 system (Illumina). RNA-seq reads were trimmed and QC-ed using Trim Galore (version 0.6.7) and read counts were identified using Kallisto (version 50.1). Differentially expressed genes were identified using the DEseq2 software package with statistical evaluation using the SciPy library.

RNA-seq data of ES cells were classified into three clusters: C1–C3. C1 genes, bound by both PRC2 (EED) and cPRC1, were poorly transcribed, whereas C2 genes exhibited less occupancy by PRC2 or cPRC1 and were more active, and C3 genes were barely bound by PRC2 or cPRC1 despite the presence of PCGF1–PRC1, but they were the most highly expressed. Upon ES to EB differentiation analysis, the expression of ES and EB genes in WT and KO was evaluated by Z-score for TPM values obtained from RNA-seq experiments, and the Z-score table for the C1 genes was used as an input vector and the C1 cluster was further decomposed into four clusters by K-mean clustering.

ChIP-seq and CUT&Tag

Chromatin from murine embryonic midbrains was collected and stored at −80°C after 1% formaldehyde fixation for 10 min and two wash steps with PBS containing a protease inhibitor cocktail (cOmplete MINI, Sigma-Aldrich). For one CUT&Tag analysis, 15 to 20 embryos were collected and used. After being thawed, tissue samples were washed once with PBS, then swelling buffer (0.1% NP-40, 1 mM dithiothreitol in PBS) was added and the mixture was incubated on ice for 10 min with occasional mixing by pipetting. After centrifugation at 8,000 rpm for 5 min, pellets were suspended in RIPA buffer, and about 3% of chromatin from human cells (HEK293T) was mixed into the samples (the exact amount of spike-in chromatin was determined by reading the input chromatin mixture) and then sheared by sonication. After sonication, the DNA concentration was measured with Qubit (Thermo Fisher), and protein A/G beads (Pierce) washed twice before use with bead-blocking buffer (0.5 % Tween 20, 0.5 % BSA in PBS) were added to the samples to minimize non-specific binding. After centrifugation at 15,000 rpm for 30 min, the supernatants were removed for the immunoprecipitation step. The protein A/G beads were washed twice with bead-blocking buffer, then suspended in bead-blocking buffer with antibody and incubated at 4° C for at least 4 h. Then, antibody-bound beads were washed twice with bead-blocking buffer and the pre-cleared chromatin samples were added. After overnight incubation at 4°C, chromatin sample–bound beads were washed with RIPA six times, RIPA-500 mM NaCl buffer twice, LiCl buffer twice, and TE buffer (10 mM Tris-HCl pH 8.0, 1 mM EDTA) twice. The antibody-bound material was eluted with elution buffer (10 mM Tris-HCl pH 8.0, 5 mM EDTA, 300 mM NaCl, 0.1 % SDS) overnight at 65°C, treated with RNase for 30 min, and then incubated with Proteinase K for 1 h at 37°C. The DNA samples were used for ChIP-seq analyses. ChIP-seq libraries were prepared by using a NEBNext Ultra II FS DNA prep kit (E7805; NEB).

ChIP-seq and CUT&Tag data processes

For the ChIP-seq experiments, reads were aligned to mm9 by using bowtie2 with the default parameters.[67] The Cut&Tag dataset, which contained a spike-in genome, was aligned against the concatenated genome of mm9 and hg38 by using bowtie2, and all uniquely matching reads were retained by using the following parameters: –no-mixed –no-discordant –very-sensitive. The resulting SAM files were then split such that reads aligning to *Mus musculus* and *Homo sapiens* were placed in separate files. The SAM files were converted to the BAM format by using Samtools.[68] The PCR duplicates were removed by using Picardtools (http://broadinstitute.Github.io/picard). The BAM files were converted to bigwig files by deepTools2 for visualization.[69] For comparison across ChIP-seq samples without a spike-in genome, the bigwig files were normalized to RPKM. For CUT&Tag with a spike-in genome, RPKM were further normalized according to normalization factors calculated as (Count of EED mapped to mm9) / (Count of EED mapped to hg38) * (Count of input mapped to hg38) / (Count of input mapped to mm9). Heatmaps were generated with deepTools2.[69] Genome browser tracks were produced by IGV_2.7.2 (https://software.broadinstitute.org/software/igv/). Read counts within intervals were calculated by using QuasR.[79]

FISH and immunoFISH

Immunohistochemistry, FISH and immunoFISH analyses were performed as described previously.[8] Briefly, paraffin sections were prepared and treated with HistoVT One (06380-05; Nacalai Tesque). After antigen retrieval steps, sections were treated with mouse monoclonal MEIS2 antibody (ab55647; Abcam) for immunoFISH, and hybridization steps directly followed antibody treatment. DNA fragments of about 7 kb were used as FISH probes, which were designed from about 10 kb fragments of target sequences in which repeat sequences were masked by using the RepeatMasker program (http://www.repeatmasker.org). FISH probes were prepared by PCR (Table S1). The probes were labeled with Alexa Fluor 488-5-dUTP (C11397; Thermo Fisher), Cy5-dCTP (PA55021; GE Healthcare) and Cy3-dCTP (PA53021; GE Healthcare). Confocal images were captured by using an Olympus microscope. The thickness of each layer of confocal images was 0.3 μm. Images were processed and distances between foci were measured by using the Volocity application (Improvision).

4C-seq

4C-seq analyses were conducted in accordance with a previous publication.[80] Briefly, after fixation of tissues with 2% formaldehyde for 10 min, samples were washed twice with PBS containing protease inhibitors (cOmplete MINI) and stored at −80°C. An average of 10 embryos was used per analysis. After being thawed, the samples were swollen as in the ChIP protocol and, after centrifugation at 8,000 rpm for 5 min, the samples were suspended in 0.5 mL of 1.2× digestion buffer; 7.5 μL of 20% SDS was added (final 0.3%) and the mixture was shaken at 37°C for 1 h. After the addition of 50 μL of 20% Triton X-100 (final, 1.8%) and subsequent shaking at 37° C for 1 h, 400 U of *Nla*III (R125L; NEB) was added and the samples were kept overnight with shaking at 37° C. Then, 40 μL of 20% SDS was added (final, 1.3%) and the mixture was incubated at 55°C for 20 min. The samples were diluted by the addition of 7 mL of 1.2× ligation buffer and, after the addition of 375 μL of 20% of TritonX-100 (final, 1%), the samples were incubated at 37°C for 1 h with shaking. After the samples had been chilled, 400 U of T4 ligase was added. This was followed by incubation at 16 °C for 5 h and storage overnight at 65°C to reverse crosslinking. After RNaseA and proteinase K treatment, DNA was purified from the samples and used as the 3C samples for the subsequent steps of the 4C-seq library construction. In short, 100 ng to 1 μg of 3C sample was digested with 10 U of *Dpn*II (R0543T; NEB) overnight, and then, after purification of the DNA, the samples were suspended in 200 μL ligation buffer and ligated overnight. After DNA purification, DNA samples were subjected to 10 cycles of PCR with primer pairs designed to detect specific sites by using primers for the *Meis2* promoter. They were then further amplified with an index primer (E7335L; NEB) to construct a 4C-seq library for subsequent NGS analysis. Data indicating the frequencies of the associations were normalized by the total count mapped on chromosome 2.

QUANTIFICATION AND STATISTICAL ANALYSIS

Analyses for LC-MS/MS measurement

In the KDM2B-protemome analysis, data-independent acquisition raw data were analyzed by using the SWATH processing function embedded in the PeakView software (Sciex). Peptides from pig trypsin and protease I precursor lysyl endopeptidase were used for retention time recalibration between runs. The following criteria were used for data-independent acquisition quantification: peptide confidence threshold, 99%; 30 ppm maximum mass tolerance; 6 min maximum RT tolerance. Multivariate data analysis was performed by using Markerview software (Sciex).

In the TUBE analysis, peptide quantification was calculated by the EncyclopeDIA algorithm[81] in Scaffold DIA.

The analyses for ChIP-seq and CUT&Tag

For ChIP-seq without a spike-in genome, CPM were calculated to conduct comparisons between samples and for CUT&Tag with spike-in DNA; they were further normalized based on normalising factors as described earlier. Bar graphs of CUT&Tag experiments are based on the tag number extracted by bedtools coverage. Those tag numbers were divided with total tag number and normalized with the spike-in factor as above. Each ChIP (or CT) score was divided by the sum of the scores within one set of experiments and multiplied by the number of samples (e.g., in the case of the CUT&Tag experiment = using chromatin samples from 9.0 DPC, 9.5 DPC and 9.5 DPC+PI; 3*{CT9.0}/({CT9.0}+{CT9.5}+{CTpi})). All statistical analyses and visualization of dot plots and scatterplots were performed by using R (v. 4.0.2). Whole-genome analyses using midbrain data were processed as follows: EED target promoters were defined by K-means clustering using deepTools3.5.1 (TSS ± 5 kb and number of clusters = 2) from EED CUT&Tag data from 9.0 and 9.5 DPC MB. Differentially expressed genes from 9.0 to 9.5 DPC MB were picked up by comparison between two groups of duplicated RNA-seq results using edgeR3.36.0 (p-value < 0.05 as threshold). Clustering of Skp1a-KO/control was made by fold change of EED binding and RNA-seq results (threshold -0.25>: <0.25). Read counts of EED CUT&Tag were calculated by bamCoverage from spike-in normalized bigwig files. In comparison between *Skp1a*-KO and *Skp1a*:*Pcgf1*-dKO, the scale factor in control for each was calculated and used for normalization. The *p* values denote statistical significance, as calculated by the HSD of ANOVA-Tukey.

The analysis of FISH

In FISH analysis, *p* values of distances were calculated by using the Wilcoxon test.